When More Observation Hurts: Robust Clarification in Ask-or-Act Under Instruction Ambiguity

CS6208 Final Presentation | 5-min solo talk

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A controlled benchmark for cost-aware clarification with both passive (action observation) and active (typed questions) channels.

The Question, and Why It Matters for Cooperative AI

Driving question

A cooperative helper has two channels for inferring a principal's goal from an ambiguous instruction:

- **Passive** — watch the principal's actions (inverse planning).
- **Active** — ask a typed clarification at cost c_q.

When should the assistant ask vs. act, and how do these two channels of evidence interact?

- Acting too early commits to the wrong goal.
- Asking too often wastes time and frustrates the principal.

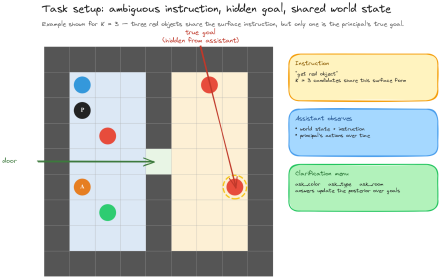
Why care (cooperative AI)

Real assistants face exactly this trade-off — but prior work studies one channel at a time:

- **CIRL** / **CLIPS**: passive inverse planning, no asking.
- **ATD** / **ShootFirst**: clarification, no action evidence.

No prior benchmark combines both channels under controlled ambiguity. We build the first one and use it to characterise channel *interaction*, not just to propose a new policy.

Setup and the AskOrAct Decision Rule



9×9 two-room gridworld, K candidate goals share the surface instruction, hidden true goal, Boltzmann principal (β, ε).

Bayes-risk one-step lookahead

$$C_{\text{Act}}(b_t) = \sum_g b_t(g)\, d^*(s_t, g)$$

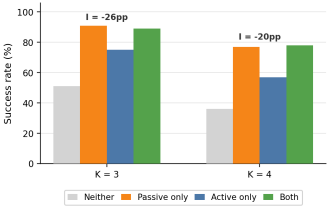
$$C_{\text{Ask}}(q) = 1 + c_q + \mathbb{E}_\alpha\big[C_{\text{Act}}(b_{t,q,\alpha})\big]$$

$$\pi = \begin{cases} \text{ask}(q^\star) & \min_q C_{\text{Ask}} < C_{\text{Act}} \\ \text{act}(\hat{g}) & \text{otherwise} \end{cases}$$

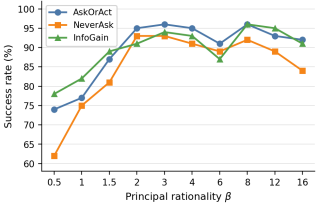
- **6 policies**: AskOrAct, NeverAsk, AlwaysAsk, InfoGainAsk, RandomAsk, POMCPPlanner.
- **Sweep**: $K \in \{1-4\}$, $\varepsilon \in \{0, .05, .1\}$, $\beta \in \{1, 2, 4\}$, 5 seeds.
- **Scale**: 19,800 episodes; 95% bootstrap CIs.

Three Diagnostic Findings (the Actual Contribution)

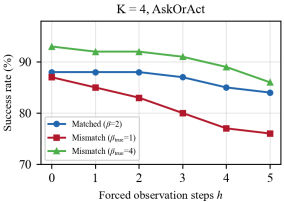
Headline: AskOrAct vs. NeverAsk **+7.9 pp** at $K{=}3$ and **+8.4 pp** at $K{=}4$ with only ∼0.6 questions/ep. The interesting story is *why*.



(1) Sub-additive channels.
$I{=}-26{\pm}3$ pp at $K{=}3$. The two channels share information; clarification matters most when passive inference is weak (+22.6 pp under full first-action aliasing).



(2) Non-monotonic in β.
NeverAsk peaks at $\beta{=}8$ (92%) and drops to 84% at $\beta{=}16$: extreme rationality collapses action diversity and hides intent. AskOrAct detects this and asks more.



(3) More observation can hurt.
Forced observation costs -0.8 pp/step matched but -2.2 pp/step under β-mismatch — passive evidence *poisons* the posterior.

Robust Clarification + Takeaways

Fixing finding (3): AskOrAct-ST

Three *decision-layer* fixes (overlap gate, lookahead, hard observation cap) all **failed** (-1 to -5 pp). $\Rightarrow$ the bottleneck is posterior quality, not the ask-or-act gate.

AskOrAct-ST adds *surprise-tempered* Bayes updates: down-weight observations whose likelihood is much lower than the model expects.

- Matched models: $+0.0 \pm 1.6$ **pp** (no harm).
- Mismatched (β_{true}=1): $+1$ to $+5$ pp.

Takeaways

1. Passive + active are **sub-additive** — design ask-policies to exploit channel *redundancy*, not stack them.
2. Assistability is **non-monotonic** in principal rationality; cost-aware asking is the corrective.
3. Under mismatch, fix the **posterior**, not the gate.

Code: `github.com/Yufannnn/AskOrAct`

AI declaration: Claude Sonnet 4.6 was used for coding, debugging, LaTeX editing, and figure generation throughout. I am responsible for all research decisions and content.